

## M04 - Feature Engineering and Feature Stores



# Topic

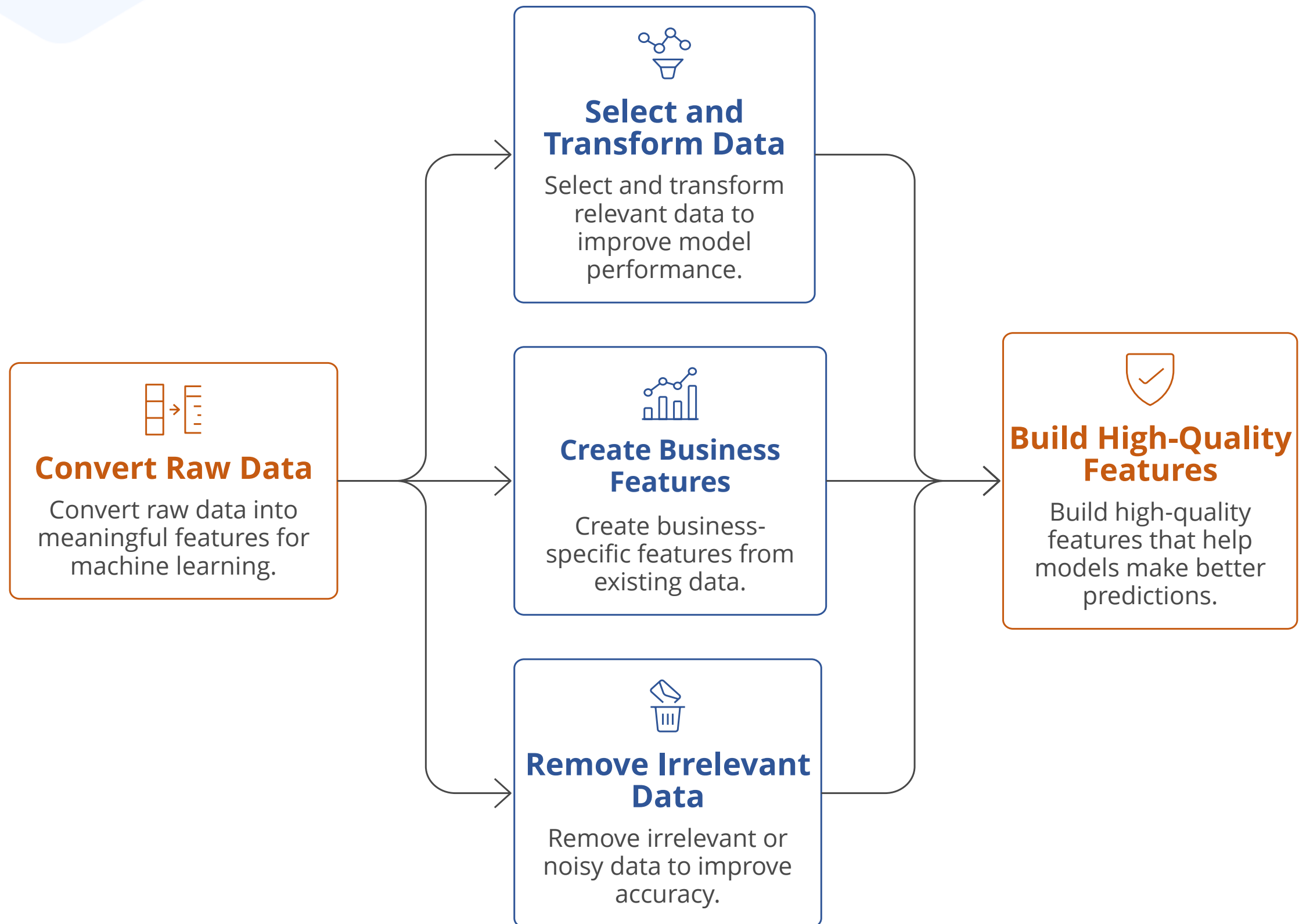
- **Introduction to Feature Engineering**
- **Feature Lifecycle Management**
- **Online vs Offline Feature Stores**
- **Feature Consistency Challenges**
- **Reusable Feature Pipelines**
- **Feature Freshness and Monitoring**
- **Integrating Feature Stores into ML Workflows**
- **Hands-on: Build and Register Features in a Feature Store**
- **Module Summary & Key Takeaways**





# **Introduction to Feature Engineering**

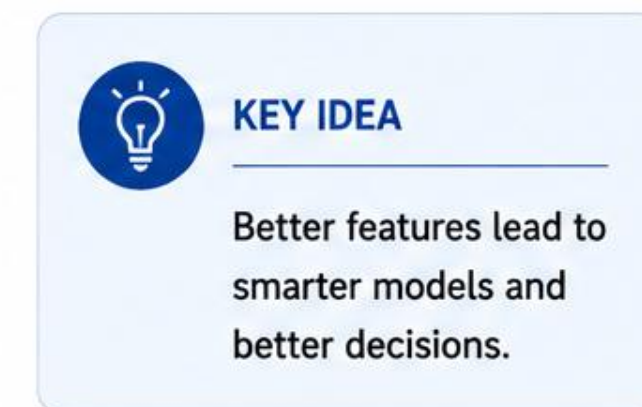
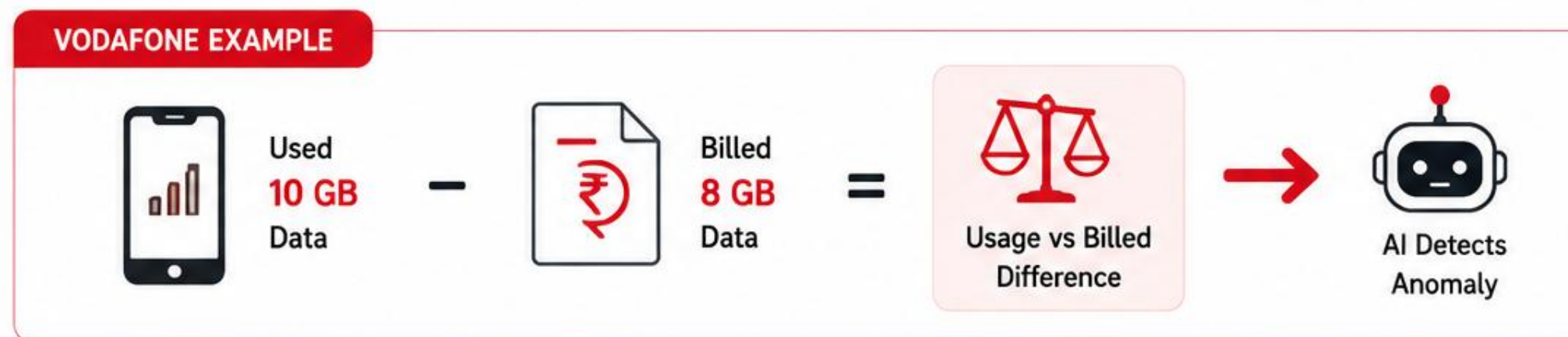
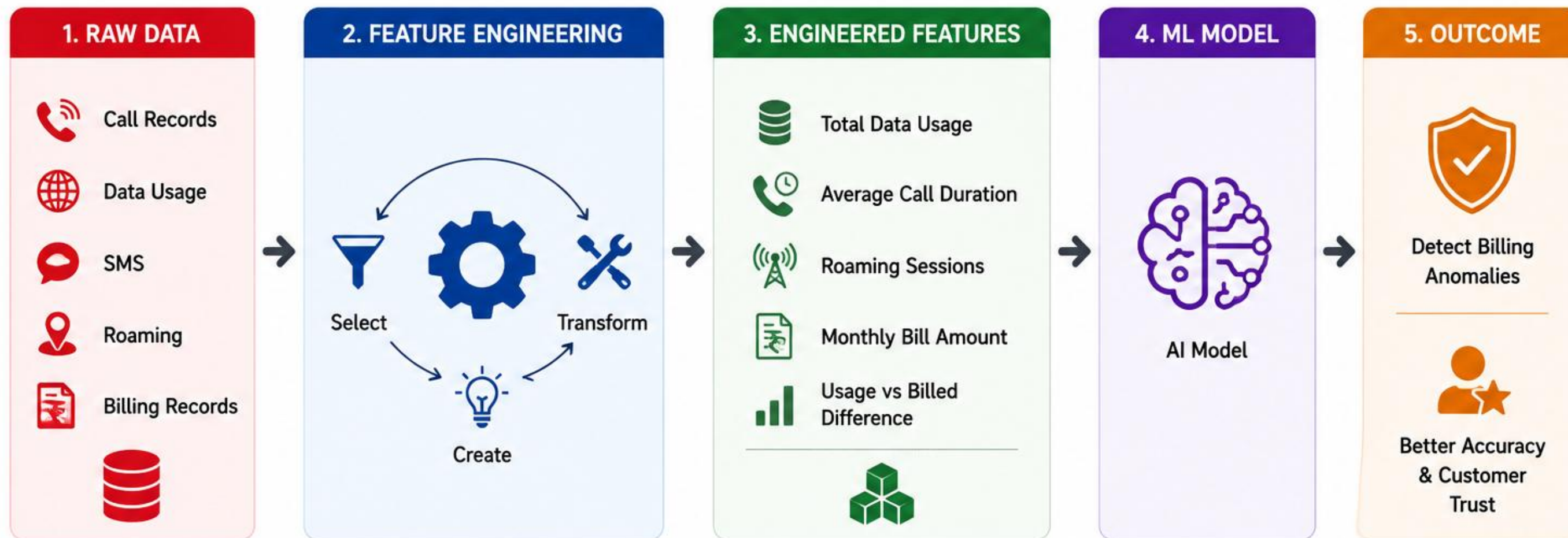
# Feature Engineering





# INTRODUCTION TO FEATURE ENGINEERING

Turning Raw Data into Smart Features for Better Predictions

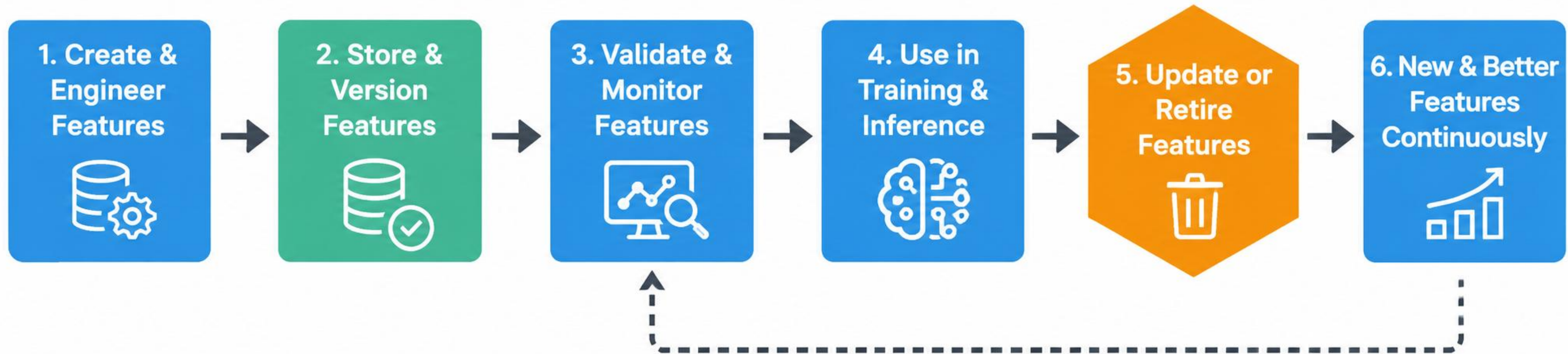


Right Features → Better Models → Better Business Impact

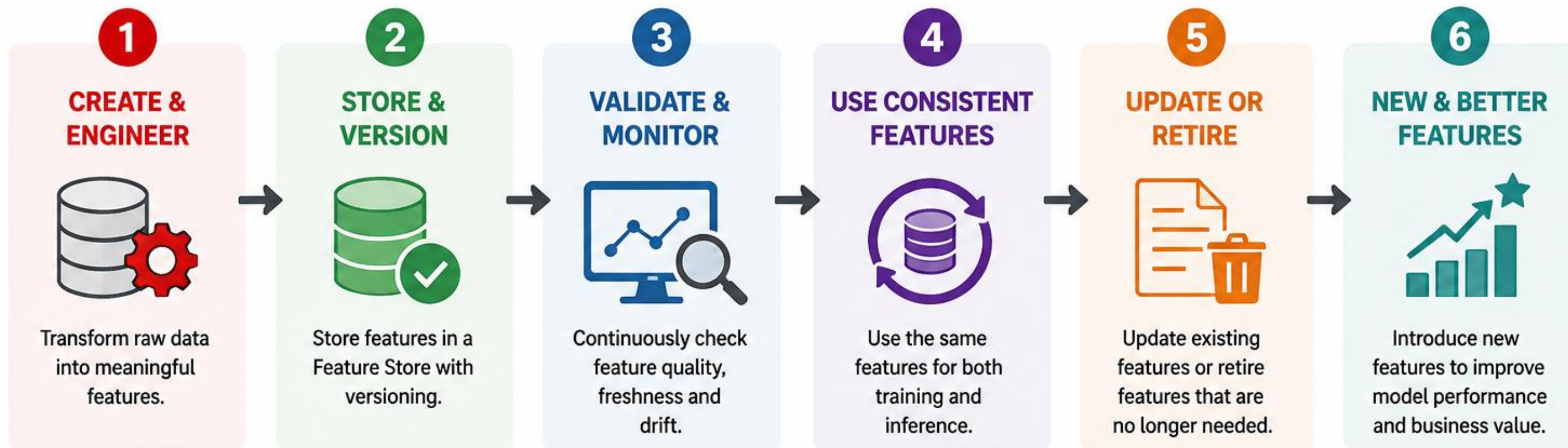


# **Feature Lifecycle Management**

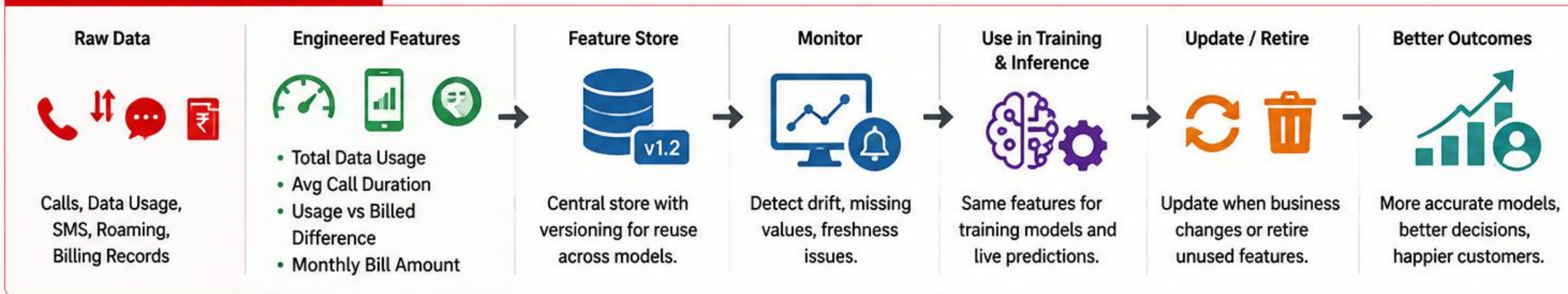
# Feature Lifecycle Management







## VODAFONE EXAMPLE – BILLING AI



Consistent Features



Better Models

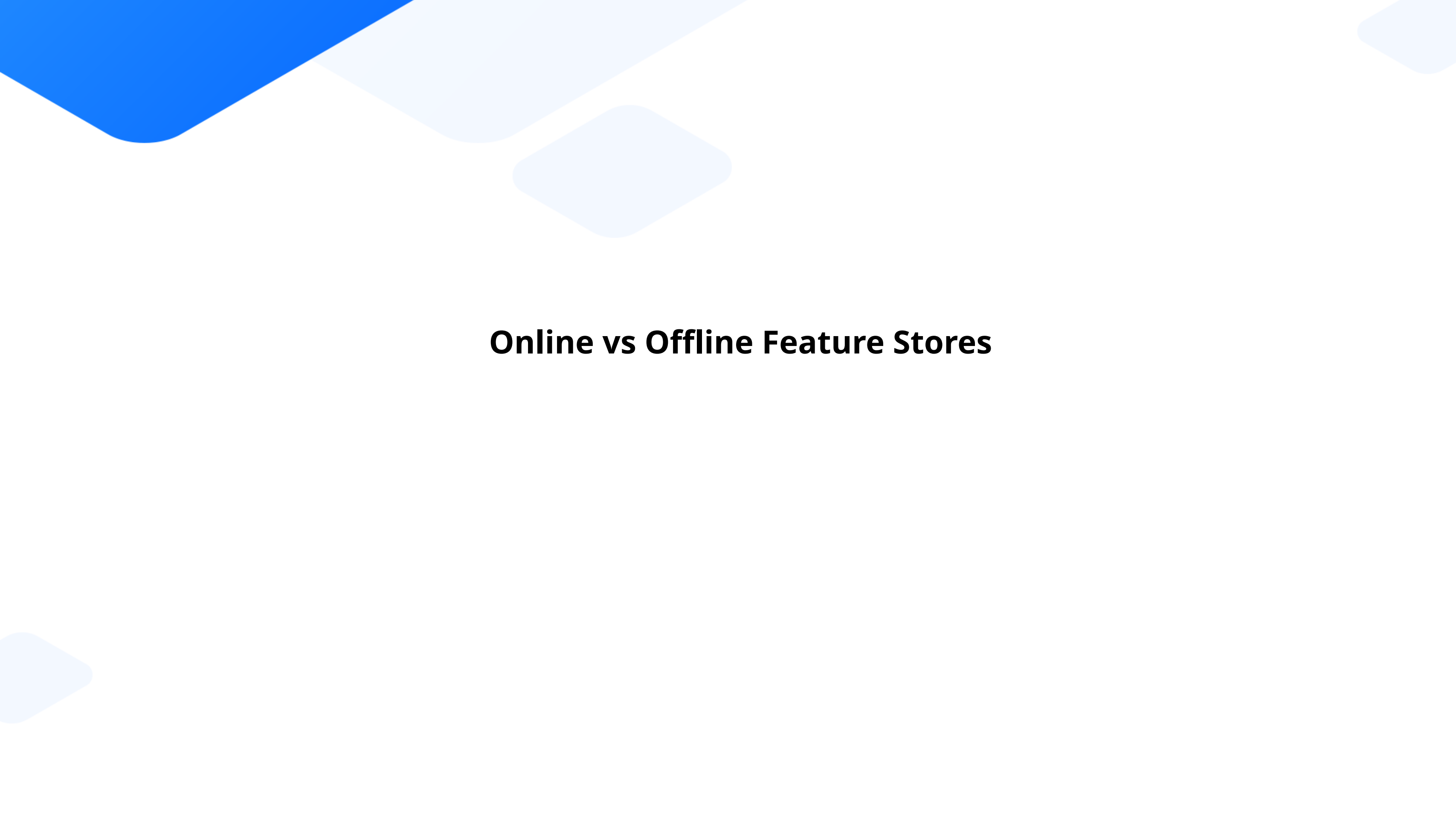


Trusted AI



Business Impact





## **Online vs Offline Feature Stores**

# Dataset Versioning Process

| Feature Store Type                                                                                              | Primary Purpose                                                       | Examples                                                            |
|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------|
|  <b>Offline Feature Store</b>  | Stores historical features for batch model training                   | Azure Data Lake, Amazon S3, Databricks, Snowflake, PostgreSQL, HDFS |
|  <b>Online Feature Store</b> | Serves the latest features with low latency for real-time predictions | Redis, DynamoDB, Azure Cache for Redis, Cassandra                   |

**Offline** means the feature store is optimized for **training** (minutes or hours are acceptable).

**Online** means the feature store is optimized for **serving predictions** (milliseconds are required).



## OFFLINE FEATURE STORE

For Batch Training & Analytics



### Purpose

Store historical features for model training and batch processing.



### Data

Large volume of historical data (days, months, years).



### Access Pattern

High throughput, not time critical (minutes or hours are OK).



### Use Cases

Model training, backtesting, analytics, reporting.



### Vodafone Example

Store 12 months of billing, usage, roaming data in Azure Data Lake to train Billing AI model.

### Examples



Azure Data Lake



Amazon S3



Databricks



Snowflake



PostgreSQL

...



## ONLINE FEATURE STORE

For Real-time Inference & Low Latency



### Purpose

Serve latest features for real-time predictions.



### Data

Latest / real-time feature values (seconds, minutes).



### Access Pattern

Low latency, high QPS (milliseconds are critical).



### Use Cases

Real-time predictions, fraud detection, alerts, personalization.



### Vodafone Example

Store customer's latest data usage and bill amount in Azure Cache for Redis to detect billing anomaly instantly.

### Examples



Redis



Amazon DynamoDB



Azure Cache for Redis



Cassandra

...

## CAN BE DEPLOYED ANYWHERE



Deployment location is independent of online/offline.

### Cloud + Offline (Example)



Azure Data Lake, Amazon S3, Snowflake

### Cloud + Online (Example)



Azure Cache for Redis, DynamoDB, Redis

### On-Premises + Offline (Example)



PostgreSQL, Hadoop, SQL Server

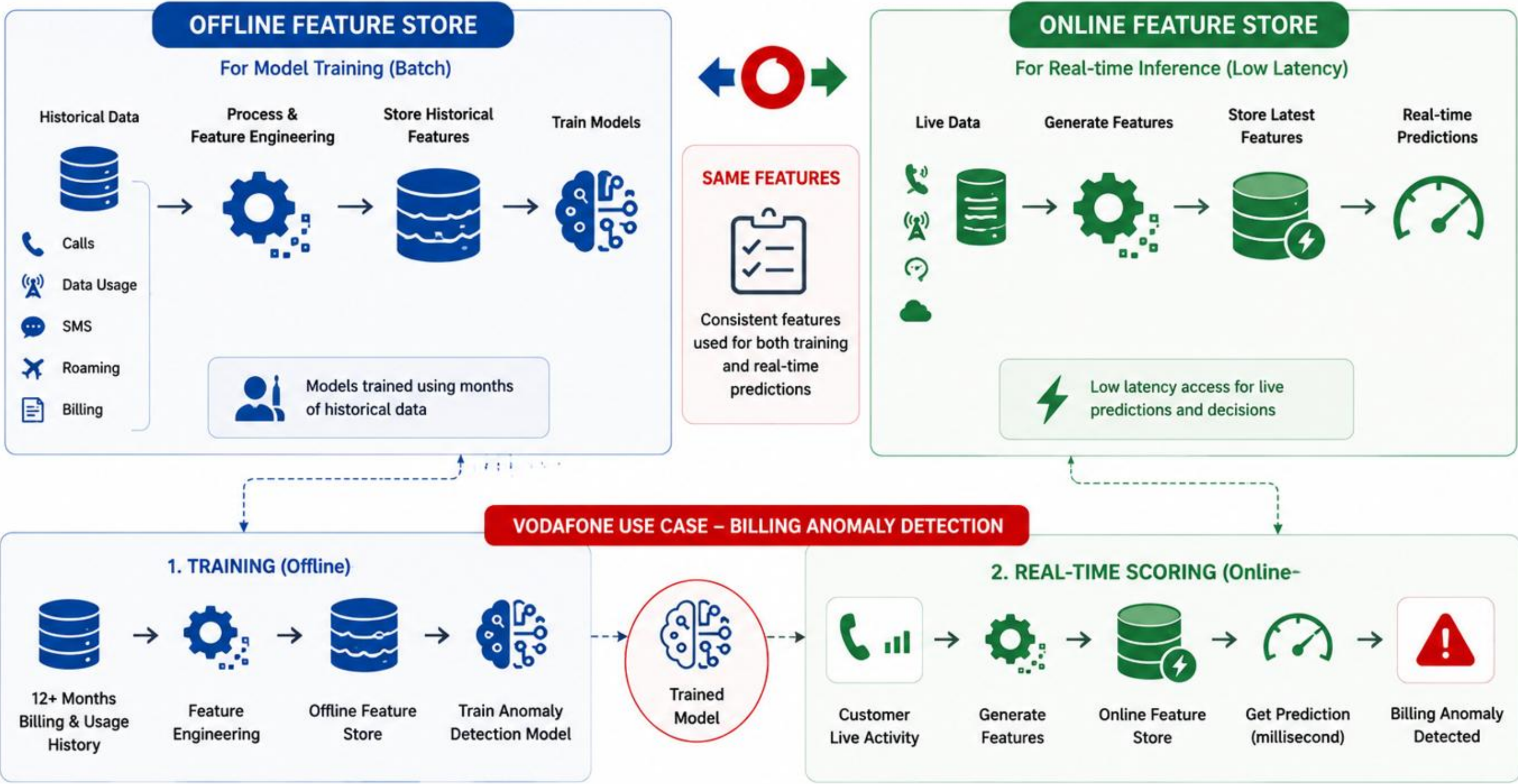
### On-Premises + Online (Example)

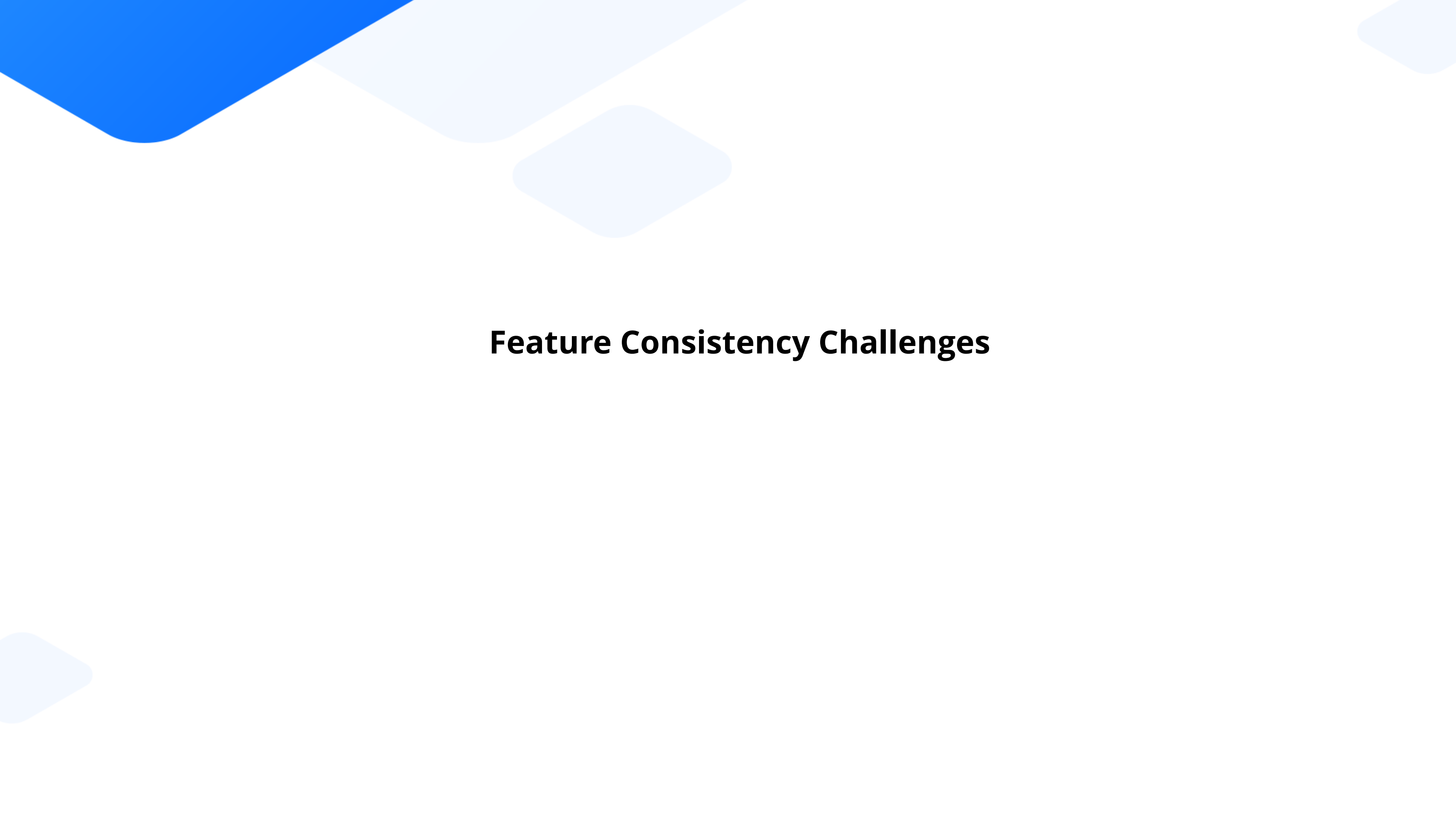


Redis, Cassandra, Memcached



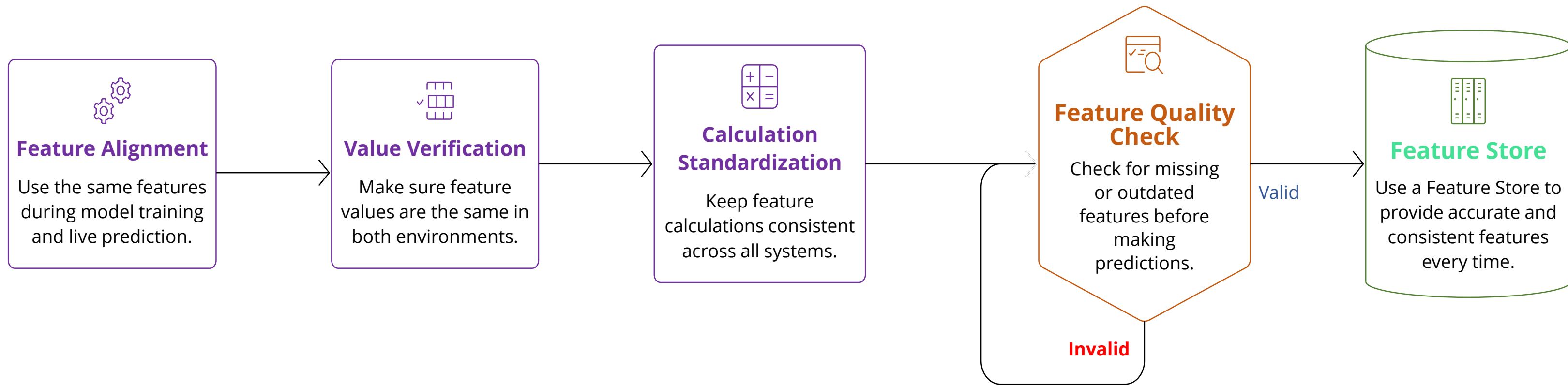
# How we can use that





## **Feature Consistency Challenges**

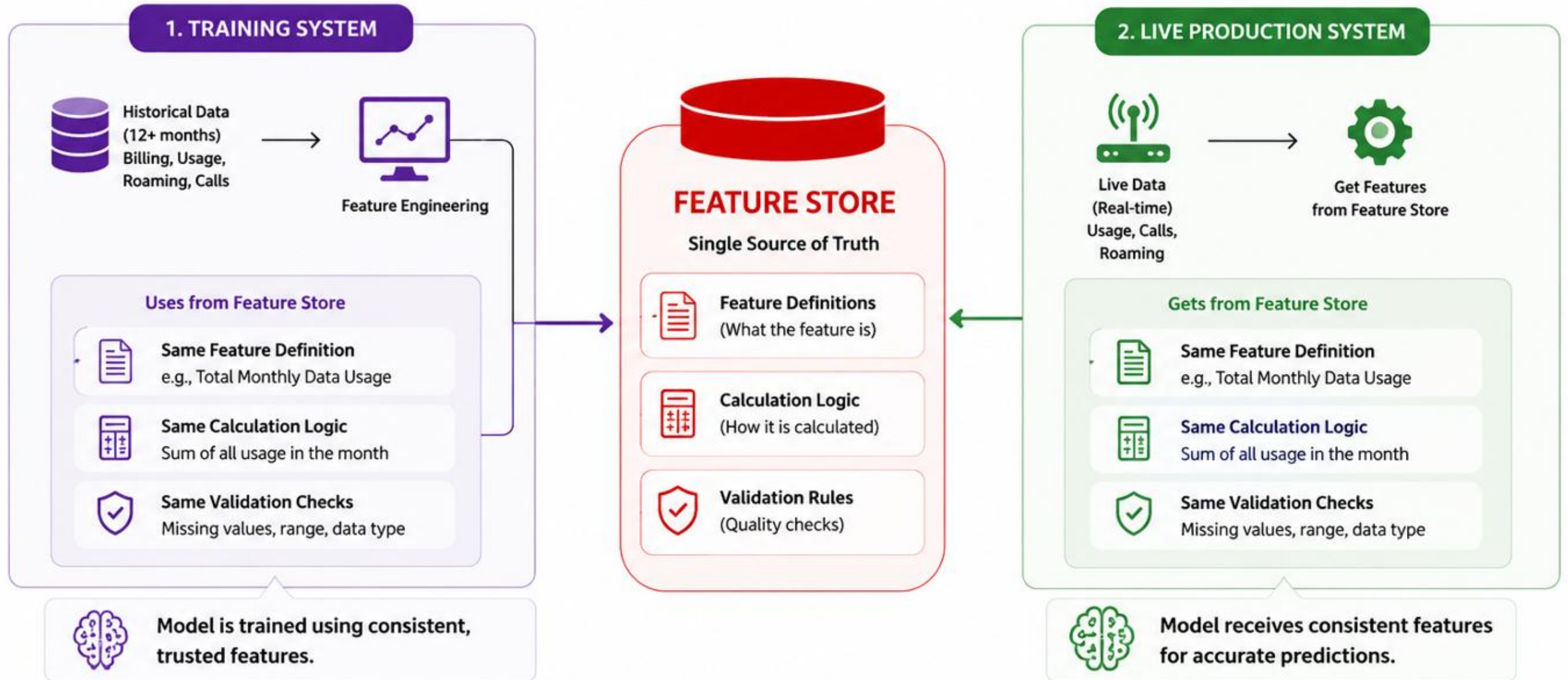
# Feature Consistency Challenges





# One Feature Store. Same Features. Same Results.

Consistent features for training and live predictions



Same features used  
in training and  
production



No training-serving  
skew

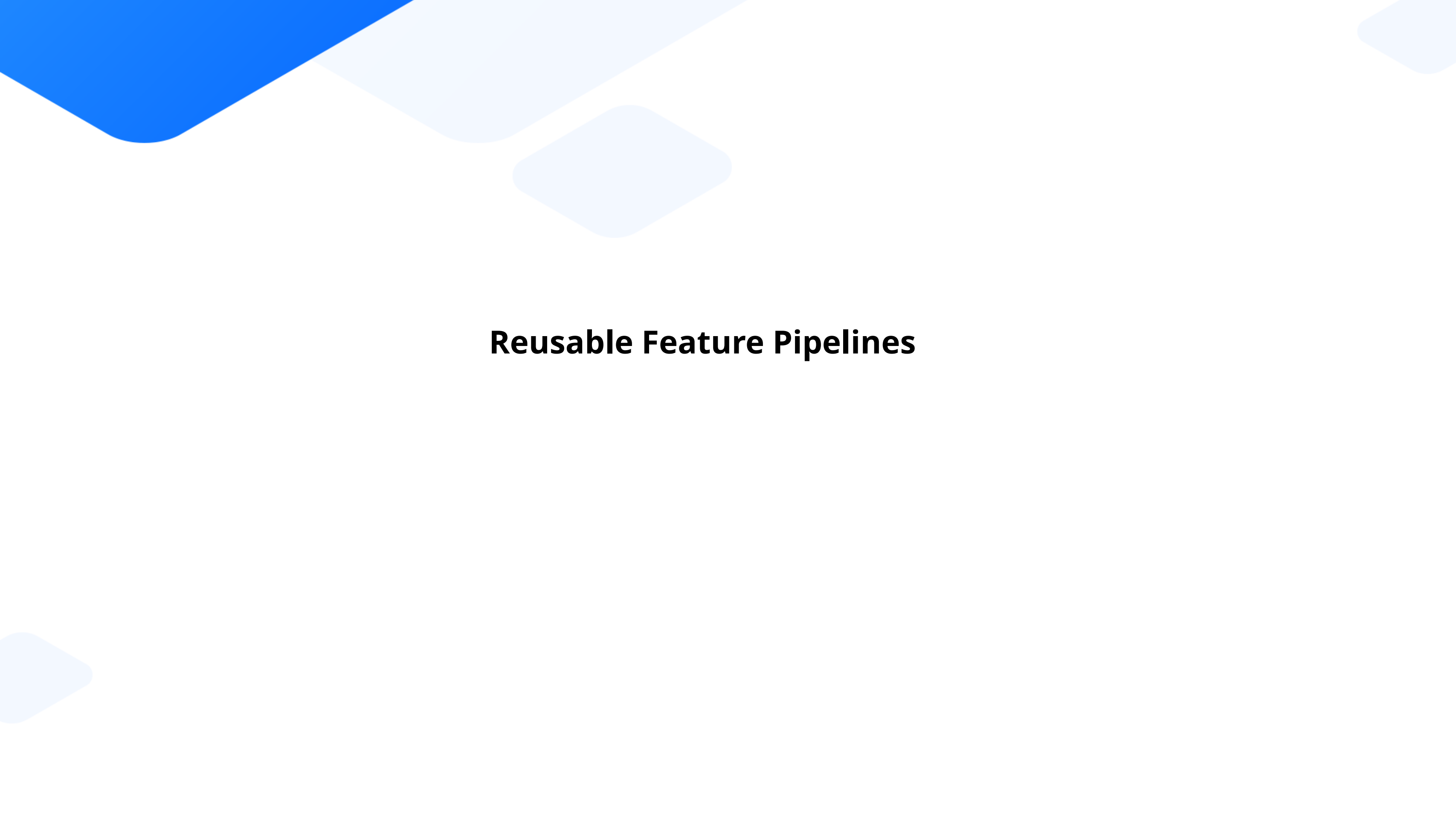


Accurate predictions  
and reliable alerts



**Better decisions.  
Happier customers.  
Zero revenue leakage.**





# **Reusable Feature Pipelines**

# Reusable Feature Pipelines

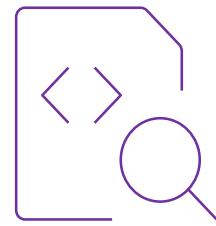
1



## Build Reusable Pipelines

Create feature pipelines once

2



## Avoid Duplication

Save development time and effort

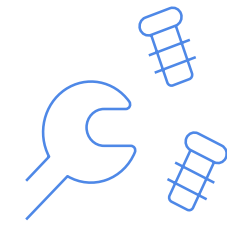
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## Ensure Consistency

Use same logic across projects

4



## Simplify Maintenance

Update pipeline in one place



## VODAFONE EXAMPLE

Multiple AI models need the same features

-  Total Monthly Data Usage
-  Average Call Duration
-  Monthly Bill Amount



## ONE REUSABLE FEATURE PIPELINE



## REUSED BY MULTIPLE MODELS



**Model 1**  
Billing Anomaly Detection



**Model 2**  
Customer Churn Prediction



**Model 3**  
Offer Recommendation Model



All models use the same features with the same logic and quality.



Consistent data leads to better predictions!

## UPDATE ONCE, IMPACT EVERYWHERE



Example: New 5G data plan changes how data usage is calculated.



Update the pipeline in one place.



All models automatically get updated features.

## BUSINESS BENEFITS



Faster Development



Consistent & Reliable Data



Lower Maintenance



Better AI Outcomes



Reusable Feature Pipelines help Vodafone deliver **consistent, high-quality** features to all AI models—**faster** and **smarter**.



## **Feature Freshness and Monitoring**

# Feature Freshness and Monitoring

## Keep features updated

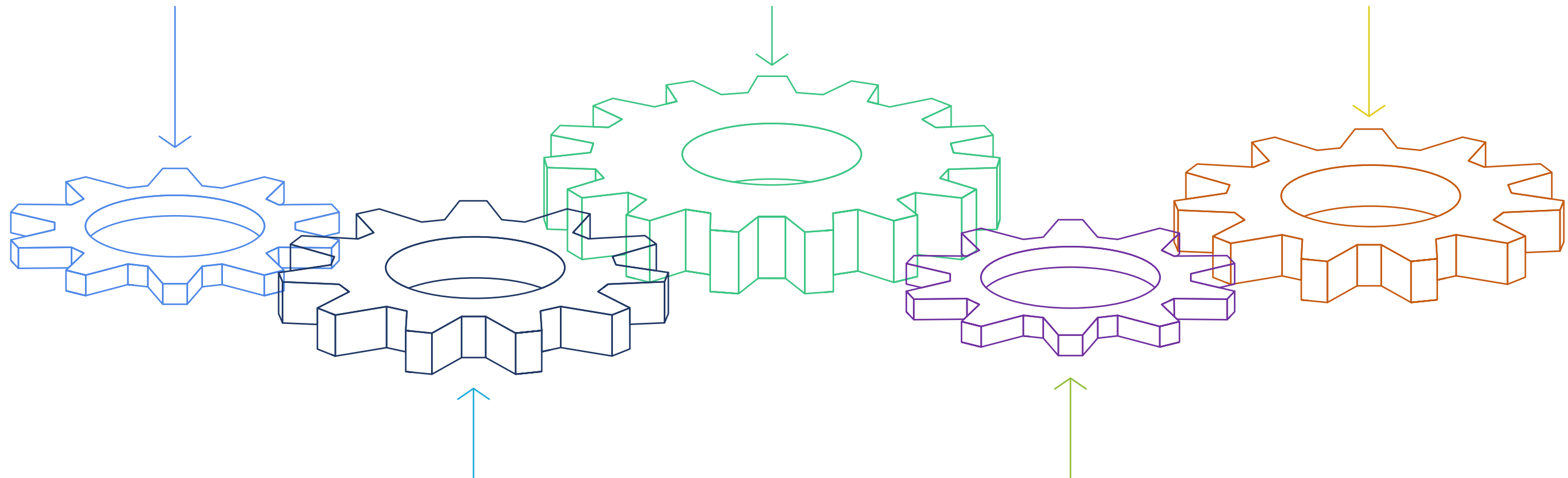
Keep features updated with the latest business data.

## Detect outdated features

Detect outdated or missing features automatically.

## Refresh features regularly

Refresh features regularly to ensure reliable AI predictions.



## Check feature values

Check that feature values are correct before making predictions.

## Monitor feature quality

Monitor feature quality to maintain model accuracy.



## Example: Vodafone Billing AI



Customer is using mobile data throughout the day

9:00 AM



Data Used  
**5 GB**

Data updates happen

5:00 PM



Data Used  
**10 GB**

If feature is not refreshed...  
.....→

AI Uses Old Value

Still shows  
**5 GB**  
(Outdated)



AI may miss billing anomaly or make wrong prediction

### 1. Data Source



Network Usage



Billing System



Customer Profile

Real-time customer data continuously generated

### 2. Feature Pipeline



Calculates features like Current Data Usage, Monthly Bill Amount, Call Duration, etc.

Runs continuously or in micro-batches

### 3. Feature Store



Always store latest feature values with timestamp

Example: Current Data Usage  
Updated at 5:00 PM = 10 GB

### 4. Monitoring & Freshness Checks



Freshness Check

Is data updated within expected time?



Completeness Check

Are any features missing?



Data Quality Check

Are values valid and in expected range?



Alerting

Notify if any issue found

### 5. AI Model (Billing AI)



Uses fresh, correct features for predictions

Detects billing anomalies accurately and reduces revenue leakage



### Without Freshness & Monitoring



Outdated feature  
(5 GB)



Wrong or missed predictions



Revenue loss & customer impact



### With Freshness & Monitoring



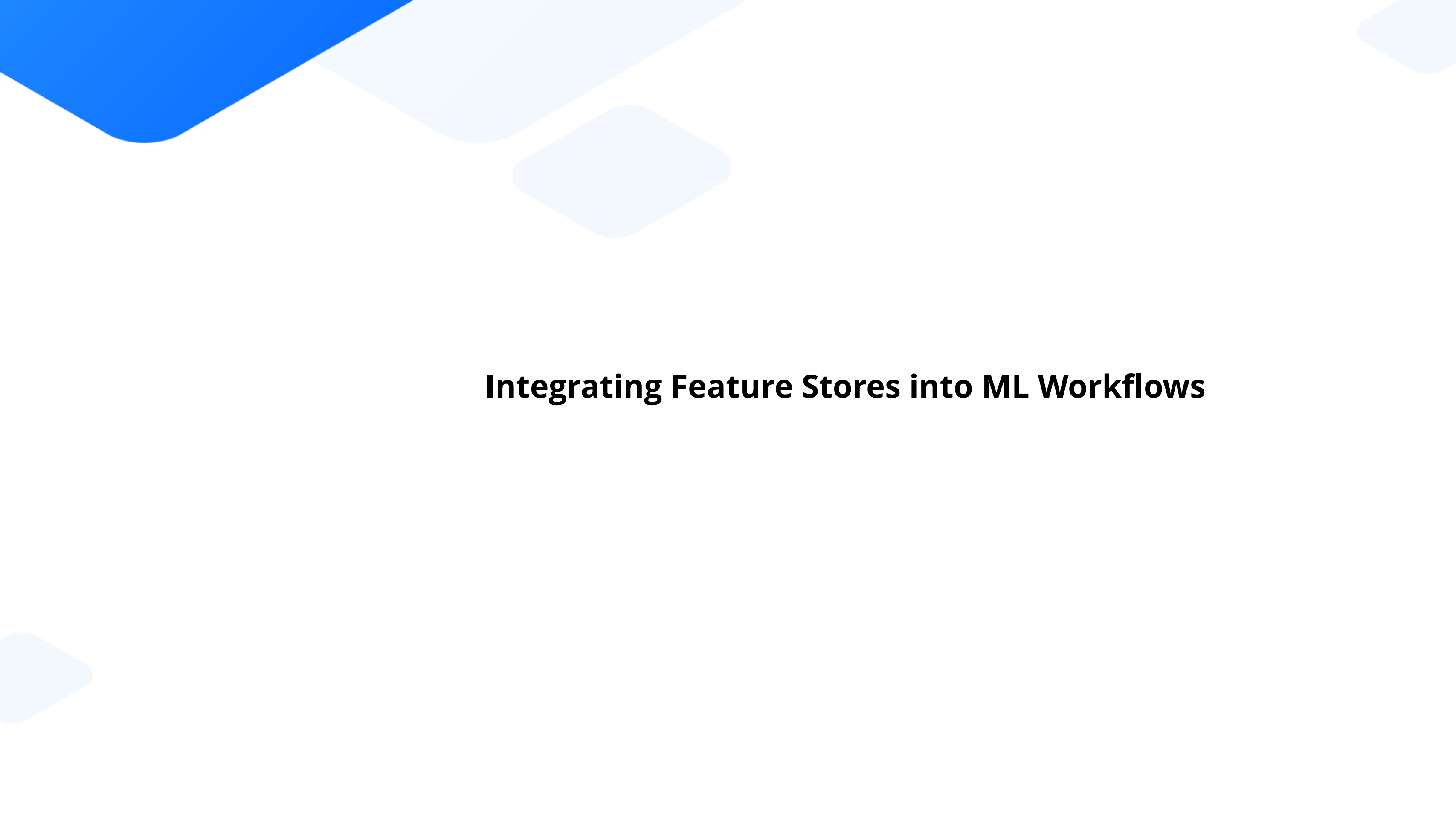
Latest feature  
(10 GB)



Accurate predictions



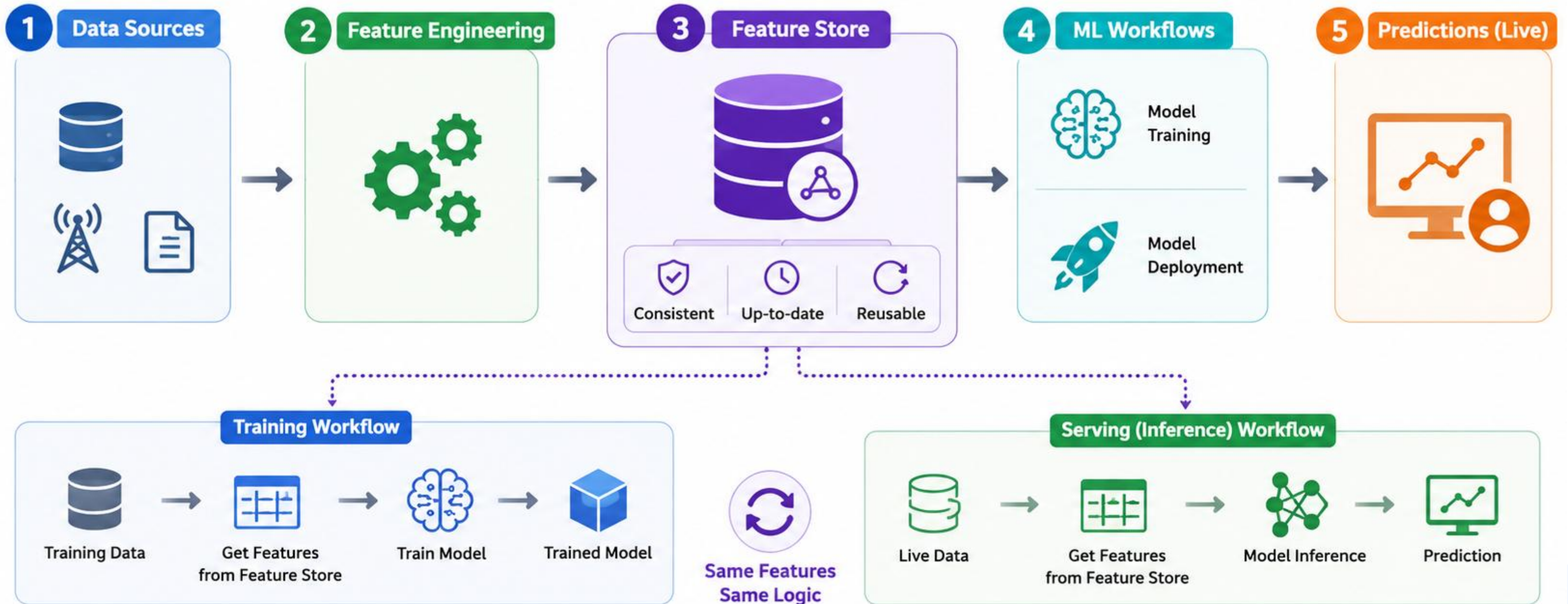
Revenue protected & happy customers



# **Integrating Feature Stores into ML Workflows**



# Integrating Feature Stores into ML Workflows





**Activity 2**

**Lab 3**



**QNA**